

BRANCH COUNTY AP, MI, USA

WMO: 725414

Lat: 41.933N

Lon: 85.053W

Elev: 292

StdP: 97.86

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-15.0	-22.0	0.5	-17.2	-18.8	0.7	-14.0	10.7	-3.8	9.4	-3.8	2.6	230	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.0	22.6	30.3	22.0	28.8	21.4	25.1	28.7	24.0	27.8	23.2	26.8	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	27.1	22.8	18.2	25.9	22.3	17.6	25.3	78.3	28.2	73.8	27.6	70.4	26.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.4	7.5	-22.5	33.7	3.1	1.7	-24.7	34.9	-26.4	36.0	-28.2	36.9	-30.4	38.2		
(5)			-22.8	26.5	2.9	1.3	-24.9	27.4	-26.6	28.1	-28.2	28.9	-30.3	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-4.1	-3.1	2.7	9.4	15.5	20.5	22.2	21.1	17.8	11.2	4.9	-1.2		
	DBStd	10.62	6.40	6.21	6.19	5.20	4.84	3.68	3.05	2.92	4.24	5.04	5.33	5.76		
	HDD10.0	1695	439	367	241	72	9	0	0	0	1	47	170	350		
	HDD18.3	3499	697	599	486	271	116	19	4	8	60	228	404	607		
	CDD10.0	1622	1	2	13	54	179	314	379	344	236	83	16	2		
	CDD18.3	384	0	0	1	3	27	83	124	94	44	6	0	0		
	CDH23.3	3244	0	0	6	29	241	712	1101	720	386	48	0	0		
	CDH26.7	932	0	0	1	4	53	222	353	178	114	7	0	0		
(14) Wind	WSAvg	3.2	3.9	3.8	3.7	3.9	3.2	2.7	2.2	2.0	2.3	3.1	3.5	3.7		
(15) Precipitation	PrecAvg	875	46	39	62	86	100	87	98	92	82	67	66	58		
	PrecMax	1126	131	135	117	170	277	174	243	236	158	213	143	116		
	PrecMin	523	5	7	11	8	26	18	17	9	1	11	8	13		
	PrecStd	134	31	26	28	37	46	42	53	54	39	39	32	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	15.9	23.7	27.1	30.5	32.9	34.0	32.3	32.2	27.7	21.1	15.9		
		MCWB	12.0	12.3	16.8	17.6	20.0	22.3	23.4	23.8	21.3	20.2	14.8	14.0		
	2%	DB	10.0	12.3	18.7	23.2	27.9	31.2	32.0	30.2	30.0	24.3	17.6	12.3		
		MCWB	8.3	10.3	14.1	15.6	19.3	21.8	23.5	22.7	21.2	17.4	13.0	10.8		
	5%	DB	7.1	8.2	15.2	21.3	26.3	29.0	30.2	28.7	27.6	22.1	15.7	8.8		
		MCWB	5.5	6.4	11.4	14.0	18.9	20.6	22.5	22.1	20.1	16.8	11.7	7.4		
	10%	DB	4.0	5.1	12.4	18.7	23.9	27.5	28.7	27.3	25.4	19.6	12.9	6.2		
		MCWB	2.7	3.2	9.0	12.8	17.6	20.6	22.0	21.5	19.1	15.3	9.8	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.8	17.3	20.4	23.1	24.8	27.2	26.6	24.5	21.6	16.3	14.1		
		MCDB	13.8	14.6	23.3	24.8	27.1	29.2	30.3	30.1	28.5	25.3	19.0	15.5		
	2%	WB	8.8	10.2	14.6	17.7	21.4	23.8	25.4	24.9	23.0	19.7	14.5	10.8		
		MCDB	9.3	11.6	17.9	20.9	25.4	28.1	28.9	28.0	26.8	22.5	17.0	11.7		
	5%	WB	5.5	6.5	12.4	15.4	20.1	22.8	24.3	23.6	22.0	17.9	12.5	7.6		
		MCDB	6.9	8.5	14.7	19.4	24.3	26.9	28.0	26.8	25.5	20.1	14.8	8.3		
	10%	WB	3.0	3.2	9.0	13.5	18.9	21.8	23.4	22.6	20.7	16.0	10.3	5.0		
		MCDB	4.0	4.8	11.7	17.9	22.6	25.8	27.0	25.7	23.4	18.8	12.7	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	9.4	11.2	10.9	10.5	10.9	10.7	11.4	10.2	8.3	6.6		
		MCDBR	9.2	10.7	13.5	14.0	12.8	12.4	12.1	12.1	13.4	13.0	11.3	9.5		
	5% WB	MCWBR	8.1	8.8	9.1	8.0	6.9	5.2	5.3	5.9	6.0	7.2	8.0	8.2		
		MCWBR	8.7	10.2	12.1	11.8	10.9	9.9	9.8	9.7	10.6	10.3	10.1	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.350	0.351	0.392	0.425	0.441	0.430	0.437	0.391	0.350	0.318	0.302		
		taud	2.373	2.193	2.333	2.243	2.195	2.191	2.230	2.203	2.315	2.425	2.462	2.405		
	Ebn at Noon		845	838	893	876	853	837	842	823	839	838	819	808		
		Edh at Noon	81	115	112	133	143	144	137	137	114	90	75	73		
(44) All-Sky Solar Radiation	RadAvg		1.46	2.33	3.52	4.61	5.34	6.02	6.11	5.29	4.31	2.79	1.75	1.21		
	RadStd		0.14	0.33	0.34	0.42	0.51	0.39	0.30	0.27	0.38	0.27	0.24	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page